Dexpirona Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-912/B

Facilities and Equipment

Manufacturing is performed at the approved commercial site using dedicated product-contact equipment trains. Equipment qualification and cleaning validation are maintained under the site validation master plan. No changes to the approved facility registration are proposed.

Deviations observed during the campaign were investigated and closed with no impact to product quality. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. The control strategy links critical quality attributes to critical process parameters identified during development.

Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Cleaning verification swab results were below the health-based exposure limits derived for the product. No changes to the approved analytical procedures are proposed in this submission. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Statistical evaluation of batch release data demonstrates process capability well within specification limits.

Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.

Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Cleaning verification swab results were below the health-based exposure limits derived for the product.

Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. No changes to the approved analytical procedures are proposed in this submission. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

Section 3.2.P.4 — Excipients

For the commercial formulation F-912/B, Plasdone K-29/32 (Povidone, binder) is used.

Grade Selection Rationale

The selection of Plasdone K-29/32 is justified by compatibility with the active ingredient demonstrated in binary stress studies observed during development studies.

Grade selection and control follow the principles of ICH Q8(R2) and USP-NF.

Current Commercial Specification

The current approved commercial specification (SPEC-EX-7927 Rev 03) lists Povidone as Kollidon 30.

Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Training records for personnel involved in manufacturing and testing are maintained under the site quality system.

No changes to the approved analytical procedures are proposed in this submission. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Deviations observed during the campaign were investigated and closed with no impact to product quality.

Container Closure System

The commercial packaging configuration consists of high-density polyethylene bottles with child-resistant polypropylene closures and induction seal liners. Container closure integrity was demonstrated by dye ingress testing. The packaging components comply with applicable food-contact regulations and compendial requirements for plastic packaging systems.

Cleaning verification swab results were below the health-based exposure limits derived for the product. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data.

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. The control strategy links critical quality attributes to critical process parameters identified during development. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Training records for personnel involved in manufacturing and testing are maintained under the site quality system.

Reference Standards

Primary reference standards for the drug substance and specified impurities are qualified against compendial standards where available. Working standards are qualified against the primary standard under an approved protocol and are stored under controlled conditions.

Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Cleaning verification swab results were below the health-based exposure limits derived for the product. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement.

Analytical Method Validation Summary

The assay and related-substances procedures employ reversed-phase high performance liquid chromatography with ultraviolet detection. Validation demonstrated acceptable specificity, linearity, accuracy, precision, and robustness. Forced degradation studies confirmed the stability-indicating capability of the related-substances method.

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. The control strategy links critical quality attributes to critical process parameters identified during development.

Process Validation Summary

Process performance qualification was executed on three consecutive commercial-scale batches. All predefined acceptance criteria for critical process parameters and critical quality attributes were met, supporting a conclusion of a validated state of control.

Statistical evaluation of batch release data demonstrates process capability well within specification limits. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. No changes to the approved analytical procedures are proposed in this submission. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection.